Scoria in the hand

A card to print, laminate and keep in the sample bag. The numbers are the ones the vault's rock notes carry; the tests are the ones that settle a sample at the outcrop.

Five tests, in order

Heft	Light for its size but it **sinks**. If it floats it is pumice, whatever the colour.
Vesicles	Rounded, 1–10 mm, walls glassy to dull. More than about 30 % of the volume is holes; under 30 % call it a vesicular basalt.
Colour	Black or dark grey when fresh; red, brown or yellow where the iron oxidised while the clast was still hot.
Crystals	Few or none visible. Olivine or plagioclase specks are fine; a glittering groundmass is not scoria.
Streak	Dark grey to brown on unglazed porcelain. A white streak means a felsic rock.

Numbers worth remembering

	ρ_bulk (g/cm³)	Φ
Scoria, typical	1.3	55 %
Scoria, dense end	1.9	32 %
Pumice	0.7	75 %
Reticulite	0.1	97 %
Basalt, parent lava	2.8	0 %

Bulk vesicularity from two densities, with the dense-rock equivalent at 2.8 g/cm³:

$$\Phi = 1 - \frac{\rho_{\mathrm{bulk}}}{\rho_{\mathrm{DRE}}}$$

Logging a sample

At the outcrop

1. Photograph the clast in place with the scale bar before touching it.
2. Note the locality, the unit it came from and whether it fell, flowed or was dug out.
3. Give it an ID: two letters for the locality, a dash, a running number (SC-01, ET-07).
4. Bag it with the ID written on the bag **and** on a tag inside.

Back at the bench

Mass in air	To 0.1 g, after a night in the drying cabinet at 60 °C.
Seal	Paraffin wax or cling film; note which, and weigh again.
Mass in water	Suspended from the balance hook, fully submerged, no bubbles on the surface.
Bulk density	Mass in air over the displaced volume, corrected for the seal.
DRE	Pycnometer on the powdered offcut, or 2.8 g/cm³ for a basalt if there is no offcut to spare.

Two readings to distrust

- A vesicularity above 80 % in a rock that sinks: the seal leaked and water got into the vesicles.
- A bulk density above 2.5 g/cm³ in a rock with visible holes: the clast is not what the bag says it is.

Scoria field guide · the card is page 1, the protocol page 2. Embedded in the vault as `![[Field card.pdf]]`.